

GOKULNATH N

Computer Vision & MLOps Engineer

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PROFESSIONAL SUMMARY

Computer Vision & MLOps Engineer with **3 years** delivering **4 production AI systems** — yard management, vehicle tracking, idle-time monitoring, and chute-mapping automation — across 3 logistics terminals with **600+ concurrent cameras**, **40% GPU reduction**, and **sub-100ms inference latency**. Skilled in YOLO/ONNX/TensorRT optimisation, Kubernetes, and Azure event-driven pipelines.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, Flask, FastAPI, REST APIs, WebSockets, Event-Driven Architecture

Computer Vision & DL: PyTorch, OpenCV, YOLO, YOLOX, ONNX, TensorRT, Deep Learning, Object Detection, Multi-Object Tracking, OCR, License Plate Recognition, Video Analytics, Model Training, Fine-Tuning, Data Augmentation, PTQ/QAT, Neural Network

MLOps & Deployment: Docker, Kubernetes, OpenShift, Kubeflow, CI/CD, GitHub Actions, MLflow, Model Registry, Experiment Tracking, Pipeline Orchestration, Model Monitoring, GPU Optimization, Pod Management, Node Allocation, Real-Time Inference, Containerization

Cloud & Database: Azure Functions, Azure Event Grid, Azure SQL, Database Design, Query Optimization

Monitoring & Tools: Grafana, Prometheus, Linux, GitHub, Bitbucket, Production Support

AI / GenAI: LLM, RAG, HuggingFace (exploring integration with CV pipelines)

PROFESSIONAL EXPERIENCE

Computer Vision & MLOps Engineer

— Tata Consultancy Services (TCS) | Client: PostNord AB

Sept 2023 – Present

- Trained and fine-tuned YOLOX object detection and multi-object tracking models on custom datasets; data augmentation improved **mAP by 15%** across varying terminal conditions.
- Optimised GPU inference via ONNX export and TensorRT, reducing **GPU utilization by 40%** and achieving **30+ FPS/stream**.
- Built Flask/FastAPI backends with WebSocket streaming processing **500+ detection events/sec**; integrated Azure Event Grid and Azure SQL.
- Containerised services with Docker; deployed on Kubernetes/OpenShift with HPA; CI/CD via GitHub Actions — **deployment time cut from days to hours**.

PRODUCTION USE CASES — POSTNORD AB

1. Yard Management System

[PyTorch](#) · [YOLOX](#) · [OpenCV](#) · [Flask](#) · [Azure SQL](#) · [WebSockets](#)

Parking Slot Detection & Allocation

2. Vehicle Entry/Exit Tracking

[YOLO](#) · [OCR](#) · [License Plate Recognition](#) · [OpenCV](#) · [Azure SQL](#) · [REST API](#)

Gate Access Automation

3. Idle Time Tracker

[PyTorch](#) · [Object Detection](#) · [OpenCV](#) · [Azure Functions](#) · [Azure SQL](#)

Storage Zone Monitoring

4. Chute-Mapping Automation (Smart Connected Terminal)

[PyTorch](#) · [YOLOX](#) · [ONNX](#) · [TensorRT](#) · [Flask](#) · [Azure Event Grid](#) · [Kubernetes](#) · [Kubeflow](#) · [MLflow](#)

Latest · Most Complex

- Detected trucks and containers from overhead cameras; mapped each vehicle to its parking slot in real time and pushed **automated slot-availability alerts** to drivers via WebSocket, eliminating manual yard monitoring.
- Detected vehicles at entry/exit gates, extracted number plates via OCR, and logged timestamped events to Azure SQL — delivering **fully automated, audit-ready gate records** with zero manual logging.
- Detected roller cages and cardboard boxes in reserved storage zones; tracked **dwelt time per object** using frame-level timestamps and surfaced idle-duration alerts, enabling data-driven parcel load/unload scheduling for terminal teams.
- Monitored conveyor-belt chutes via camera; detected parcel accumulation and **auto-triggered worker reallocation rules** based on chute status conditions — eliminating manual supervisor walkdowns and fully automating shift-clearing assignments.

KEY ACHIEVEMENTS

- 40% GPU reduction** across 600+ production camera streams via ONNX export and TensorRT inference optimisation.
- 4 end-to-end AI systems** delivered in production — yard management, vehicle tracking, idle-time monitoring, and chute-mapping automation — across 3 terminals with 99%+ uptime.
- Eliminated manual supervisor walkdowns** by automating chute-clearing worker assignments via rule-based AI detection pipeline.
- Zero manual gate logging** achieved through fully automated vehicle entry/exit tracking with OCR and license plate recognition.
- Sub-100ms inference latency** at 30+ FPS/stream sustained across 600+ concurrent cameras using Kubernetes HPA and GPU optimisation.

CERTIFICATIONS & EDUCATION

Certifications: Python for CV with OpenCV & Deep Learning (Udemy) · Modern CV & Deep Learning with PyTorch (Udemy)

Education: **B.Tech — Information Technology** Adhiyamaan College of Engineering | 2019–2023 | CGPA: 8.7 / 10.0